



NAME: _____ ID: _____

SCHOOL: _____ MUNICIPALITY: _____

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gas' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero degrees Celsius	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs Energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First-order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Drop in vapor pressure	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of a sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of a sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of a circle	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Long. of a circumf.	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3		4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003	
2	3 Li 6.941	4 Be 9.012												5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305												13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956		22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906		40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	*	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	**	104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

CODE: _____

Day 1. Grade 11th. Questions 2, 5, 7 and 8.



NAME: _____ ID: _____

SCHOOL: _____ MUNICIPALITY: _____

Question 2. Chemistry is for all! (18% of total; only 11th grade) Day 1

2.1	2.2	Total	Total
40	32	72	18

One day, the *Captain*, as Naylen is affectionately called, decided to share a coffee with some pre-selection students in the courtyard of her house: Leticia, Melissa, Cynthia, Jennifer, Lauren, Amanda, Gabriela, Britany, Brenda and Diana. She was very upset because she had heard her son Renny say that chemistry was “a man’s

thing”. So, together they devised a substance-identification challenge to test the knowledge of the boys’ peers. The next day, they checked the school’s reagent warehouse, where technician Ivis was waiting for them and provided them with some substances. Next, they went to the laboratory, where mentor Agustín had gathered the boys.

“Good morning, teacher,” said the *Captain*. “The ladies on the team bring a challenge to the gentlemen.”

“Whoever does not know how to identify the substances, does not eat the cake that is in the snack that the mothers have prepared,” said Agustín to motivate them. And the parade of substances began.

i. Leticia presented a white solid. “My substance is a ternary salt, whose anion is a halide and can be used as fertilizer, because plants can absorb the cation. It is very soluble in water, resulting in a slightly acidic solution. When heated, it breaks down into two pungent and toxic odorous gases, which quickly recombine when the temperature drops, causing a white fog.” Leticia dissolved a portion in water, and when she added a few drops of AgNO_3 , a white solid precipitated, which dissolved slightly upon adding NaOH to the test tube.

ii. Melissa also brought a flask with a white solid in it. “My substance is a ternary salt; it is used as disinfectant and bleach. It dissolves easily in water and feels soapy to the touch because it is basic. When it reacts with hydrochloric acid, it releases a yellowish-green gas with a very pungent odor. It has an intense yellow coloration in the flame. In solution, it reacts with excess KI , causing an intense yellow-brown coloration.” Melissa added a portion of H_2O_2 , and an intense bubbling of one odorless gas occurred, causing the flame of a match brought close to the lid of the container to intensify.

iii. “Well, I carry a quaternary substance, also white, which is used as a medicine to treat indigestion and in the kitchen as a substitute for baking powder. The aqueous solution of my substance turns bluish green a strip of universal pH indicator paper,” Cynthia said. “My substance dissolves in water, acids, and strong bases. When dissolved in acid, it causes an effervescence capable of extinguishing the flame of a match when it is brought closer to the test tube. In solution, it reacts with calcium chloride, forming a white turbidity.” Cynthia performed the flame test with a combustion spoon, and an intense yellow coloration was observed.

iv. Jennifer brought a flask with a black powder. “My substance is elementary; it occurs in various allotropic forms. The one I have here allows me to write and is also used as a lubricant.” Next, she took portions of the solid and stirred vigorously with water, an acid solution, a basic solution and ethanol; however, no dissolution occurred in any case. When a lit match was brought into contact to a portion of the solid, it burned vigorously. In a test tube under an extraction hood, Jennifer placed the unknown substance and black copper(II) oxide and heated them strongly, obtaining a reddish residue.

v. Lauren showed a colorless crystalline solid. “My substance is a molecular acid, ternary, it exists in the form of dihydrate and is slightly toxic in the body, causing the formation of kidney stones; it is also used as a primary standard in acid-base titrations.” Lauren heated a portion in a test tube, vapors formed that partially condensed on the edge of the tube and extinguished the flame of a match. When an excess of its aqueous solution was added to acidulated KMnO_4 , gradual discoloration was observed that accelerated over time; the discoloration was faster when the experiment was repeated under heating.

CODE: _____

Day 1. Grade 11th. Questions 2, 5, 7 and 8.



NAME: _____ ID: _____

SCHOOL: _____ MUNICIPALITY: _____

vi. Amanda brought a solid yellow color with a pungent smell. "My substance is elemental; it comes in the form of two main allotropes composed of cyclic molecules. It is insoluble in water, but soluble in carbon disulfide. If it burns in the presence of oxygen, it gives off a gas with a pungent odor, which, when bubbling in an orange solution of acidified $\text{K}_2\text{Cr}_2\text{O}_7$, turns it green; moreover, the same gas instantly discolors a violet acidulated solution of KMnO_4 ." Amanda heated a portion of the solid with concentrated NaOH and then carefully added drops of HCl . The rotten-egg-like odor was immediately detected, and turbidity was observed in the test tube.

vii. Gabriela showed a white crystalline solid. "This substance is ternary and is used in pyrotechnics due to its explosive nature. It dissolves in water with difficulty, and the resulting solution is neutral. Its cation causes a violet-lila coloration to the flame. When heated strongly, it decomposes, leaving a white residue and releasing a gas that fuels the combustion of burning splitter." Gabriela added a portion of concentrated hydrochloric acid to the substance, and a yellowish-green, pungent-smelling gas was released.

viii. The twin Britany brought a crystalline, white solid. "The substance inside the flask is ternary; it is widely used in agriculture as a fertilizer, also as an explosive and in instant cooling packages. Its aqueous solution is slightly acidic. When it is strongly heated, it decomposes, releasing gases that can make a wood chip burn more intensely, producing a characteristic brownish smoke." To identify the cation and anion present, Britany conducted two separate assays. First, she took a portion of the solid and added it to a test tube containing a concentrated KOH solution. When heated, a colorless gas was released, turning a wet strip of universal pH paper blue. Then, when a few drops of iron(II) sulfate solution were added to the aqueous solution, followed by concentrated sulfuric acid, a lower layer formed, and a brown ring appeared at the interface between the two liquids.

ix. The other twin, Brenda, brought a white solid as well. "This substance is the most abundant on the beach and in deserts. It is very hard, has a high melting point and is used to make glass. It does not dissolve in water, organic substances, acids or weak bases at room temperature, but it does dissolve in HF ." Brenda added a portion to a test tube with concentrated sodium hydroxide and strongly heated the solution. A small portion began to dissolve.

x. "And now it's my turn!" Diana said, wearing a pair of gloves. She then took what looked like a small, grayish ball from inside a flask and carefully threw it into the bucket of water that Ivis had discreetly placed nearby. And kaboom! The instant the ball entered the water, a yellow flame was observed inside the bucket, and an explosion occurred, spilling all the water and soaking some of the students. Diana quickly pulled out a toy gun loaded with phenolphthalein (HX) solution and began spraying some of the boys, causing their clothes to stain pink.

From such a loud noise, Director Dunierky immediately went up angrily to put things in order. "Gentlemen, what in the world is going on here?" Until he saw half the students soaking wet and the other half's faces red with laughter. "I knew that this was some invention of the old man... You'd better leave everything very clean, or I'll send you to field work. And do not make any more noise because in the school there is a visit from the province." Then he went on his way.

2.1. Help the boys on the team identify the substances that each of the girls had.

2.2. Write down the adjusted chemical equations of all the chemical reactions described.

Note: Please do not attempt to replicate these experiments at your school or at home.

CODE: _____

Day 1. Grade 11th. Questions 2, 5, 7 and 8.



NAME: _____ ID: _____

SCHOOL: _____ MUNICIPALITY: _____

Question 5. Thermochemistry of NaPF₆ (10% of total; only 10th and 11th grades) Day 1

5.1	5.2	5.3	5.4	5.5	5.6	Total	Total
4	2	2	2	18	12	40	10

The hexafluorophosphate anion (PF₆⁻) forms salts that can be used as electrolytes in lithium-ion and sodium batteries. The thermal and chemical stability of these salts is a critical factor in energy storage technologies. Sodium hexafluorophosphate can be obtained by reacting phosphorus pentachloride, sodium chloride and hydrofluoric acid.

5.1. Represent the Lewis structure of the anion. Say the hybridization of phosphorus and the FPF angle.

5.2. Pose the equation of the salt formation reaction NaPF₆.

The lattice energy is a measure of the stability of ionic solids and can be calculated from the Kapustinskii equation below. *U* is the lattice energy in kJ/mol, *r* and *z* are the radii in pm, and the ions' charges, respectively. $\sum v$ is the total number of ions in the empirical formula.

$$U \approx -1,202 \cdot 10^5 \cdot \frac{\sum v \cdot |z^+||z^-|}{r^+ + r^-} \cdot \left(1 - \frac{34,5}{r^+ + r^-}\right)$$

5.3. Propose an expression to estimate the radius of the PF₆⁻ ion from the covalent and ionic radii.

5.4. Estimate the lattice energy of NaPF₆ if the thermochemical radii of the ions are 102 and 242 pm.

5.5. Use the thermodynamic data **in the table** (energies in kJ/mol, 298 K) and the Born-Haber cycle to calculate the formation enthalpy of NaPF₆. Consider that each P₄ molecule has six P-P bonds.

Sublimation energy of Na(s), Δ <i>H</i> _{sub} (Na)	+108
Primary ionization energy of Na(g), <i>I</i> ₁ (Na)	+496
Sublimation energy of P ₄ (s), Δ <i>H</i> _{sub} (P ₄)	+59
Bond dissociation energy of P-P(g), <i>E</i> (P-P)	+205
Bond dissociation energy of F-F(g), <i>E</i> (F-F)	+155
Energía bond energy of P-F in PF ₆ (g), <i>E</i> (P-F)	+490
Electron affinity energy of the radical PF ₆ (g), Δ <i>H</i> _{AE} (PF ₆)	-770
Lattice energy of NaPF ₆ (s), <i>U</i> (NaPF ₆)	-600
Lattice energy of NaF(s), <i>U</i> (NaF)	-910
Primary electron affinity energy of F(g), <i>A</i> ₁ (F)	-328

5.6. Calculate the enthalpy for the thermic decomposition of the complex salt.



CODE: _____

Day 1. Grade 11th. Questions 2, 5, 7 and 8.



NAME: _____ ID: _____

SCHOOL: _____ MUNICIPALITY: _____

Question 7. Back to eleventh Grade (12% of total; only 11th and 12th grades) Day 1

7.1	7.2	7.3	7.4	7.5	7.6	Total	Total
6	8	10	8	8	8	48	12

7.1. Estimate at what temperature the sea water freezes; given that the mean salinity is 3.5% in mass, the mean density is 1.025 g/mL. Suppose that the only marine salt is NaCl. $K_{cr}(H_2O) = 1.86\text{ }^{\circ}\text{C}\cdot\text{kg}\cdot\text{mol}^{-1}$

7.2. In blast furnaces, where metal is produced from the reaction of ore (oxide) with carbon, several equilibria coexist. Three possible reactions of **one mole** of dioxygen with carbon and carbon monoxide are known. Estimate at what temperature the reduction of the mineral with carbon, forming monoxide, is the most thermodynamically favored reaction among the possible.

Substance	O ₂	C	CO	CO ₂
ΔH_f°	0	0	-111	-394
S°	205	6	198	214

7.3. You want to prepare 1.00 L of aqueous phosphate buffer solution of pH = 7.5 and total molar concentration equal to 0.50 mol/L. An aqueous solution of commercial phosphoric acid (~15 mol/L) and solid sodium hydroxide is available for this purpose. Which are the two species of phosphorus that predominate at this pH value? What amounts do you need to mix of each reagent? For H₃PO₄ we know that pK_{a1} = 2.15, pK_{a2} = 7.20 and pK_{a3} = 12.35.

7.4. Most chemical elements form hydroxides and sulfides that are poorly soluble in water. In the 2+ oxidation state, the vast majority of sulfides are more insoluble than hydroxides. In the 3+ and 4+ oxidation states, on the other hand, some sulfides exist only in the solid phase and are rapidly hydrolyzed to form hydroxides in the presence of moisture; for example, this occurs for iron(III) and aluminum. Suppose you have an aqueous solution of MnCl₂ 10⁻⁵ mol/L. Estimate the concentration of sodium sulfide required to initiate precipitation. What product is formed? $K_{ps}(\text{Mn}(\text{OH})_2) = 2.0\cdot 10^{-13}$, $K_{ps}(\text{MnS}) = 3.0\cdot 10^{-10}$; for H₂S we know pK_{a1} ~ 7.0 and pK_{a2} ~ 13.0.

7.5. Calculate the potential of the silver-silver chloride reference electrode Ag(s)|AgCl(s)|KCl(sat.) with respect to the normal hydrogen electrode, knowing that $K_{ps}^{\circ}(\text{AgCl}) = 1.77\cdot 10^{-10}$, $E^{\circ}(\text{Ag}^+/\text{Ag}) = 0.7996\text{ V}$ and that a saturated solution of KCl contains approximately 4.81 mol/L of salt. Note that in such a concentrated solution, the ions interact strongly, and their "effective concentration" or activity is only a factor or fraction of 0.588 (mean ionic activity coefficient) with respect to the molar concentration.

7.6. A green dihydrogen production plant uses an electrolyzer consisting of 200 cells. Each cell operates at a current of 5.00 A and requires 1.85 V. The performance or faradaic efficiency of the process is 98.5%, that is, this is the fraction that participates in the desired electrochemical reaction. Calculate the mass of dihydrogen produced each hour and the energy consumption.

CODE: _____

Day 1. Grade 11th. Questions 2, 5, 7 and 8.



NAME: _____ ID: _____

SCHOOL: _____ MUNICIPALITY: _____

Question 8. Ouro Preto (15% of total; 11th and 12th grades) Day 1

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	8.10	8.11	8.12	8.13	Total	Total
2	8	4	4	2	2	4	3	8	2	9	2	10	60	15

Last year, Cuba participated in the Mendeleev International Olympiad for the second time in Belo Horizonte, Brazil. The “picadillo team” was made up of Juan Carlos (called *Yepe* by everyone), Roy Roger, Marco Antonio, Ernesto Alejandro and Professor Julieta, who is also president of the Cuban Society of Chemistry. The result obtained was very relevant, with a silver medal and three bronze medals. During the event, the team visited the city of Ouro Preto, notorious for gold mining in the late 17th and early 18th centuries. The teacher, who loves chemistry very much, took the opportunity to remind the students of the electrochemical properties of this element.

8.1. Ernesto was struck by the fact that gold is widely used in electronics due to its easy malleability, ductility and excellent electrical conductivity. Why do the triad Cu, Ag and Au have the highest conductivity of all the elements in the periodic table?

8.2. *Yepe* learned that the city of Ouro Preto received that name because the gold ore found in the area was obscured by a mixture of oxides (approximately 28% and 30%, by mass of oxygen) of a certain transition element. Can you find out which element it is and the formula of its oxide?

Julieta really likes jewelry and working with metals. As she is inquisitive, she found out from the organizers what the composition of the gold medals awarded during the event was and the details of the coating process, and she also received one golden medal as a gift.

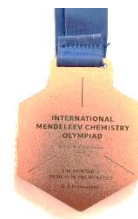
The medals were made of brass (Cu-Zn alloy), weighed approximately 210 g and had 5.0 μm of gold plating. The gold coating was performed in a solution of $\text{K}[\text{Au}(\text{CN})_2]$ 0.040 M, at pH = 4.0, using as stabilizer and buffer a solution of 75 g/L of citric acid (H_4Cy , 192 g/mol), at 50 °C, at a voltage of 2-4 V, with a current density of 0.50 A/dm² and a gold sacrificial anode. The faradaic efficiency of the process is close to 85%.

8.3. Gold(III) cyanide does not exist, but when the two ions come into contact in solution, a pseudohalogen is released, and a precipitate is formed. Formulate the reaction and explain why it happens.

8.4. Au(I) is unstable in solution, but can be stabilized by halide ions, cyanide, thiocyanate, and thiosulfate. Why? Calculate the disproportion constant of Au(I) in water at 25 °C.

Yepe wanted to know the exact composition of the medals. He determined that the mass of the medal that Julieta was given was 210 g and measured its dimensions, obtaining 40 mm side length and 6.0 mm thick. Using data available online, he was able to determine the composition but did not reveal it to his colleagues.

8.5. Assume for the rest of the question that the medal is shaped like a regular L-sided hexagon, with an area equal to $A = \frac{3\sqrt{3}}{2} \cdot L^2$. Calculate the approximate density of the alloy.



CODE: _____

Day 1. Grade 11th. Questions 2, 5, 7 and 8.



NAME: _____ ID: _____

SCHOOL: _____ MUNICIPALITY: _____

8.6. The volume obtained by *Yepe* is approximate, because the medal was actually irregularly shaped. Explain a more accurate method of measuring volume in the laboratory.

8.7. The efficiency of the gold coating process is often affected because collateral half-reactions occur at the anode. One reaction releases gas and another reaction forms a precipitate that passivates the electrode; thus, requiring higher operating voltage. Formulate the balanced equations of these two reactions.

Roy then said that his father is a jeweler and that, in his workshop, he uses aqua regia (a mixture of HNO_3 and HCl , in 1:3 proportion) to dissolve certain pieces, because acids alone cannot do so. Julieta explained that this reaction is possible due to the formation of the complex $[\text{AuCl}_4]^-$.

8.8. Help Roy calculate the standard Au^{3+}/Au reduction potential in the absence of chlorides.

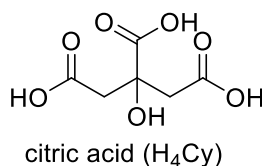
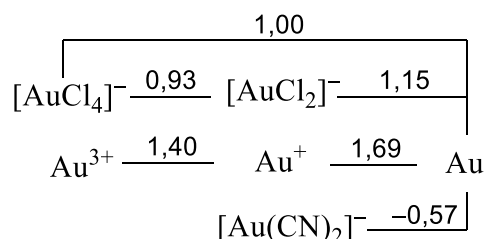
8.9. Formulate the possible reactions of gold with H^+ , NO_3^- and O_2 in the air, in the presence of chlorides. Determine which is thermodynamically favored, assuming a medium with $\text{pH} = 0$ and 1.0 M chloride.

8.10. In practice, the reaction you found above is not the one that happens. Propose an explanation.

8.11. When Marco calculated the amount of gold (19.3 g/cm^3) in grams consumed in a medal, he was surprised by the result. How long does it take to coat a gold medal?

8.12. At the end of the tour, the teacher, a specialist in organic compounds, asked the students why one of citric acid's constants is extremely small. What would you have answered?

8.13. A 2.00 g sample of filings from a gold medal was dissolved in hot nitric acid. The concentrated solution was cooled, filtered and neutralized with care to avoid precipitation of the metal hydroxides. Then, it was diluted until it completed a liter of solution in a volumetric flask. A 10 mL aliquot was transferred to an Erlenmeyer bottle, and a small amount of distilled water and 1 g of potassium iodide were added. The resulting suspension was quickly titrated with a 0.0100 M thiosulfate standard, using starch as an indicator. The volume of titrant required to achieve the first blue coloration was 21.1 mL. What is the total composition (in percent by mass) of the IMChO gold medal?



$K_{\text{ps}}(\text{AuCN}) = 1,0 \cdot 10^{-38}$ $E^\circ(\text{V}): \text{O}_2/\text{H}_2\text{O} = 1,23; \text{NO}_3^-/\text{NO} = 0,96$

$\text{pK}_a(\text{H}_4\text{Cy}) = 3,13; 4,76; 6,40; 14,4$ Air: 79% N_2 and 21% O_2 (v/v), 1,0 atm

Note: The author declares that no IMChO medal was damaged during the preparation of this question.

CODE: _____

Day 1. Grade 11th. Questions 2, 5, 7 and 8.